Day-2 Notes Computer Network Notes

Types of Networking Devices + Physical Layer + Data Link Layer + Network Layer

---

1. Introduction to Computer Networks

A **computer network** is a collection of interconnected devices that communicate with each other to share data, resources, and services. These devices may include computers, laptops, mobile phones, printers, servers, routers, switches, and many more.

The main purpose of a network is:

- data sharing
- resource sharing
- communication
- remote access
- internet connectivity

Whenever two or more devices exchange data, networking concepts such as **addressing**, **routing**, **frames**, **packets**, and **protocols** come into play.

To understand how communication happens, we study:

- **networking devices**
- **OSI layers**
- **functions of each layer**

In these notes, the focus is on:

- **networking devices**
- **Physical Layer**
- **Data Link Layer**
- **Network Layer**

These three layers form the base of actual communication in a network.

---

2. Networking Devices Overview

A networking device is a hardware component used to connect computers or networks and help data move from one place to another.

Different devices operate at different layers of the OSI model.

Some important networking devices are:

- Repeater
- Hub
- Bridge
- Switch
- Router
- Gateway
- Modem
- Access Point
- NIC
- Firewall

Each device has a different role. Some only pass signals, some forward frames, and some route packets between networks.

---

3. Classification of Networking Devices by Layer

Devices at Physical Layer

- Repeater
- Hub
- Modem
- Cables / Connectors
- NIC (partly Layer 1)

Devices at Data Link Layer

- Bridge
- Switch
- NIC
- Wireless Access Point

Devices at Network Layer

- Router
- Layer 3 Switch
- Some Firewalls

Devices operating on multiple layers

- Gateway
- Firewall
- Proxy server

---

4. Repeater

A **repeater** is a networking device used to regenerate or amplify weak signals so they can travel a longer distance.

When signals travel over a cable, they become weak due to attenuation. A repeater receives the weak signal, cleans it, strengthens it, and sends it again.

Main function

It extends the communication distance.

Layer

Physical Layer (Layer 1)

Characteristics

- works only on signals
- does not understand frames or packets
- does not check addresses

- does not filter traffic

Example

If two devices are far apart in a LAN and the cable length is too large, a repeater can be placed in between to strengthen the signal.

Key point

A repeater does not know where the data is going. It only boosts the signal.

---

5. Hub

A **hub** is a simple networking device that connects multiple devices in a LAN.

When a hub receives data from one device, it sends that data to **all devices** connected to it. It does not know the correct destination.

Layer

Physical Layer (Layer 1)

Main function

It acts like a central connection point for multiple devices.

Working

Suppose 5 computers are connected to a hub. If computer A sends data to computer B, the hub forwards the data to all 5 computers. Only the intended computer accepts it, others ignore it.

Characteristics

- no intelligence
- no MAC address table
- no filtering
- one collision domain

- half duplex communication

Disadvantages

- wastes bandwidth
- creates collisions
- not secure
- slower than switch

Key point

A hub broadcasts incoming data to all connected devices.

---

6. Bridge

A **bridge** is a device used to connect two LAN segments and filter traffic using MAC addresses.

Unlike a hub, a bridge is smarter. It checks the destination MAC address and decides whether to forward the frame or not.

Layer

Data Link Layer (Layer 2)

Main function

It divides a LAN into segments and reduces unnecessary traffic.

Working

A bridge learns which MAC addresses belong to which segment. If the source and destination are on the same segment, it does not forward the frame. If they are on different segments, it forwards it.

Characteristics

- uses MAC address

- filters traffic
- reduces collisions
- more intelligent than hub

Advantage

Improves network performance by reducing unnecessary frame forwarding.

Key point

A bridge connects LAN segments and forwards frames selectively using MAC addresses.

---

7. Switch

A **switch** is one of the most important networking devices in modern LANs. It connects multiple devices and sends data only to the intended device.

Layer

Usually **Data Link Layer (Layer 2)**
Some advanced switches can also work at **Layer 3**

Main function

It forwards data using **MAC addresses**

Working

A switch maintains a **MAC address table**. When it receives a frame:

- it checks the source MAC and learns which port it came from
- it checks the destination MAC
- if destination is known, it forwards only to that specific port
- if unknown, it floods the frame

Characteristics

- intelligent device
- one collision domain per port
- better bandwidth utilization
- full duplex communication possible
- reduces network congestion

Advantages over hub

- sends data to correct device only
- fewer collisions
- better performance
- more secure than hub

Example

In an office LAN, all computers are usually connected through a switch.

Key point

A switch works with MAC addresses and forwards frames only where needed.

---

8. Router

A **router** is a device that connects different networks and forwards data using IP addresses.

Layer

Network Layer (Layer 3)

Main function

To route packets from one network to another.

Working

A router reads the destination IP address in a packet and decides the best path to forward it. It uses routing tables and routing protocols.

Example

Your home Wi-Fi router connects your local network to the internet.

Characteristics

- uses IP address
- connects different networks
- separates broadcast domains
- can perform NAT, DHCP, firewall functions in practical systems

Advantage

Without routers, devices in one network cannot communicate with devices in another network.

Key point

A router uses logical addressing, not physical addressing, to send packets between networks.

---

9. Gateway

A **gateway** is a device that connects networks using different protocols or architectures.

Layer

Can operate at **multiple layers**, often upper layers as well

Main function

Protocol conversion

Example

If one network uses one protocol and another uses a different one, a gateway translates between them.

Key point

A router forwards packets between similar IP networks, but a gateway can translate between completely different systems.

---

10. Modem

A **modem** stands for **modulator-demodulator**.

It converts:

- digital signals to analog
- analog signals to digital

Layer

Physical Layer

Main function

Enables data communication over telephone lines or certain ISP media.

Key point

A modem is used for internet connectivity and signal conversion.

---

11. NIC (Network Interface Card)

A **NIC** is the hardware that allows a device to connect to a network.

Layer

Works at **Physical + Data Link Layer**

Main functions

- sends and receives signals
- provides MAC address
- helps device connect to network

Important point

A NIC has a **MAC address**, also called the **physical address**.

Key point

NIC is the interface through which a computer joins the network.

---

12. Access Point

A **wireless access point** connects wireless devices to a wired network.

Layer

Mainly **Data Link Layer**

Main function

It provides Wi-Fi access.

Key point

It acts like a wireless bridge between devices and the network.

---

13. Firewall

A **firewall** is a security device that filters traffic according to rules.

Layers

Can work at **Layer 3, Layer 4, and Layer 7**

Main function

To allow or block traffic based on IP, port, protocol, or application rules.

Key point

A firewall is mainly for security, not just forwarding.

---

14. OSI Model Context for These Layers

The OSI model has 7 layers:

1. Physical
2. Data Link
3. Network
4. Transport
5. Session
6. Presentation
7. Application

In this chapter we are focusing on the bottom 3:

- Physical
- Data Link
- Network

These layers are very important because actual data movement begins here.

---

15. Physical Layer

The **Physical Layer** is the first and lowest layer of the OSI model. It deals with the actual transmission of raw bits over a communication medium.

Main responsibility

To move raw binary data (0s and 1s) from one device to another through physical media.

What it handles

- cables
- connectors
- electrical signals
- optical signals
- radio signals
- bit transmission
- data rate
- voltage levels
- transmission mode

Data unit

Bits

Devices working here

- Repeater
- Hub
- Modem
- Cables
- Connectors

Functions of Physical Layer

1. Bit transmission

It sends raw bits from sender to receiver.

2. Physical characteristics

Defines cable type, connector type, pin layout, etc.

3. Data rate control

Defines how fast bits are sent.

4. Transmission mode

It supports:

- simplex
- half duplex
- full duplex

5. Topology support

Defines how devices are physically connected:

- bus
- star
- ring
- mesh

6. Signal representation

Defines how binary data is converted into electrical or optical form.

---

16. Important Concepts in Physical Layer

a) Bit

A bit is the smallest unit of data:

- 0
- 1

b) Bandwidth

Bandwidth means the data-carrying capacity of a channel.

c) Transmission Media

The path through which signals travel.

Guided media

- twisted pair cable
- coaxial cable
- fiber optic cable

Unguided media

- radio waves
- microwaves
- infrared

d) Attenuation

Weakening of signal over distance.

e) Noise

Unwanted disturbance in signal.

f) Data rate

Number of bits transmitted per second.

---

17. Physical Layer Transmission Modes

Simplex

Communication happens in one direction only.

Example:

- keyboard to computer
- TV broadcast

Half Duplex

Communication happens in both directions, but one direction at a time.

Example:

- walkie-talkie

Full Duplex

Communication happens in both directions at the same time.

Example:

- telephone call

---

18. Physical Topologies

Bus topology

All devices share one common cable.

Star topology

All devices connect to a central hub or switch.

Ring topology

Devices form a circular structure.

Mesh topology

Devices are directly connected to many or all others.

Tree topology

Combination of star and bus.

---

19. Limitations of Physical Layer

The Physical Layer only handles transmission of raw bits. It does not know:

- who is sending
- who should receive
- whether data is correct
- whether the destination is in same network or different network

That is why higher layers are needed.

---

20. Data Link Layer

The **Data Link Layer** is Layer 2 of the OSI model. It is responsible for **node-to-node delivery** of data over the same local network.

It converts raw bits into **frames** and ensures the data can move correctly between directly connected devices.

Data unit

Frame

Devices

- Bridge
- Switch
- NIC
- Wireless AP

Main responsibilities

- framing
- physical addressing
- error detection
- flow control
- access control

---

21. Functions of Data Link Layer

1. Framing

The Data Link Layer groups bits into meaningful units called **frames**.

A frame contains:

- source MAC address
- destination MAC address
- control fields
- data
- error check information

2. Physical addressing

This layer uses **MAC addresses**.

MAC address is the physical/hardware address of a device's NIC.

Example:
`00:1A:2B:3C:4D:5E`

3. Error detection

It detects transmission errors using methods like CRC.

If a frame is corrupted, the receiver can detect it.

4. Flow control

It controls the speed of data transfer so that sender does not overwhelm receiver.

5. Access control

If multiple devices share the same medium, this layer decides who can send data and when.

Example:
Ethernet rules

---

22. MAC Address and Data Link Layer

The MAC address is very important at this layer.

What is MAC address?

A MAC address is a unique hardware address assigned to a network interface card.

Why is it used?

To deliver frames within the same local network.

Important point

MAC addressing is used only for **local delivery**, not end-to-end delivery across the internet.

---

23. Same Network Communication

Suppose computer A and computer B are in the same LAN.

If A wants to send data to B:

- A knows B's IP address
- A uses ARP to find B's MAC address
- Data Link Layer creates a frame
- source MAC = A's NIC MAC
- destination MAC = B's NIC MAC
- switch forwards the frame to B

This is **same network communication**.

---

24. Different Network Communication

Suppose computer A wants to send data to a server on another network.

Then:

- A creates a packet with destination IP = server IP
- but on local LAN, it cannot directly send to the server's MAC
- it sends the frame to the **router's MAC address**
- router then forwards packet toward destination network

Important concept

For remote communication:

- IP decides the final destination
- MAC decides the next local hop

---

25. Data Link Layer Sublayers

The Data Link Layer is often divided into two sublayers:

1. LLC (Logical Link Control)

Handles flow control and error control.

2. MAC (Media Access Control)

Handles physical addressing and media access.

---

26. Switching at Data Link Layer

A switch operates at Layer 2.

How switch works

- learns MAC addresses from incoming frames
- stores them in MAC table
- forwards frames intelligently

Benefits

- less traffic
- fewer collisions
- efficient LAN communication

---

27. Error Control in Data Link Layer

If data gets damaged during transmission, the Data Link Layer can detect it.

Common technique:

- CRC (Cyclic Redundancy Check)

This layer usually detects errors. In some protocols, retransmission support may also be provided.

---

28. Flow Control in Data Link Layer

Flow control ensures sender does not send data too fast for receiver.

This prevents:

- data loss
- buffer overflow
- congestion at link level

---

29. Access Control in Data Link Layer

If many devices use the same medium, access control is needed.

Examples:

- CSMA/CD in old Ethernet
- CSMA/CA in wireless networks

This decides when a device can transmit.

---

30. Network Layer

The **Network Layer** is Layer 3 of the OSI model. It is responsible for delivering data from source to destination across multiple networks.

Data unit

Packet

Main responsibilities

- logical addressing
- routing
- path determination
- packet forwarding
- fragmentation

Device

- Router

---

31. Main Functions of Network Layer

1. Logical addressing

This layer uses **IP addresses**.

An IP address identifies a device logically in a network.

Example IPv4:
`192.168.1.10`

Example IPv6:
`2001:db8::1`

2. Routing

The Network Layer decides which path data should follow.

3. Packet forwarding

It forwards packets from one router to another until destination network is reached.

4. Internetworking

It connects different networks together.

5. Fragmentation

If a packet is too large for a network medium, it may be broken into smaller parts.

---

32. IP Address

An IP address is a **logical address** assigned to a device.

Why called logical?

Because it can be changed by software or network configuration.

Types

- IPv4
- IPv6

---

33. IPv4

IPv4 stands for Internet Protocol version 4.

Address size

32 bits = 4 bytes

Format

Decimal notation

Example:
`192.168.1.1`

Total possible addresses

`2^32 = about 4.3 billion`

---

34. IPv6

IPv6 stands for Internet Protocol version 6.

Address size

128 bits = 16 bytes

Format

Hexadecimal notation

Example:
`2001:0db8:85a3:0000:0000:8a2e:0370:7334`

Why introduced?

Because IPv4 addresses were getting exhausted.

---

35. IPv4 vs IPv6

IPv4

- 32-bit
- 4 bytes
- limited addresses
- decimal format

IPv6

- 128-bit
- 16 bytes
- huge address space
- hexadecimal format

---

36. Routing

Routing means selecting the best path for sending packets from source to destination.

This is done by routers.

Router checks

- destination IP address
- routing table
- next hop
- outgoing interface

---

37. Routing Table

A routing table is a table maintained by the router that tells:

- destination network
- next hop

- outgoing interface
- metric/cost

Based on this table, router forwards packets.

---

38. Packet Forwarding Example

Suppose a laptop in Delhi wants to access a server in Mumbai.

Step 1

Laptop creates data.

Step 2

Transport layer adds TCP/UDP info.

Step 3

Network layer adds:

- source IP
- destination IP

Now it becomes a **packet**.

Step 4

Data link layer adds:

- source MAC
- router MAC

Now it becomes a **frame**.

Step 5

Switch forwards frame to router.

Step 6

Router checks destination IP and forwards to next network.

Step 7

At every hop, MAC addresses change, but IP addresses usually remain same from source to destination.

This is a very important interview point.

---

39. Difference Between Physical, Data Link, and Network Layer

Physical Layer

- sends raw bits
- no addressing logic
- works with signals
- devices: repeater, hub

Data Link Layer

- sends frames in same LAN
- uses MAC address
- handles node-to-node delivery
- devices: switch, bridge

Network Layer

- sends packets across networks
- uses IP address
- handles routing
- device: router

---

40. Bit, Frame, Packet Difference

Bit

Smallest unit of data
Used at Physical Layer

Frame

Bits grouped with MAC info
Used at Data Link Layer

Packet

Frame payload with IP info
Used at Network Layer

---

41. Same LAN vs Different LAN

Same LAN

Uses MAC-based delivery through switch.

Different LAN

Uses IP-based routing through router.

Important line

MAC gives local delivery, IP gives global delivery.

---

42. Broadcast Domain and Collision Domain

Collision Domain

Area where packet collisions can happen.

- Hub: one large collision domain
- Switch: one collision domain per port

Broadcast Domain

Area where broadcast frames reach.

- Switch: same broadcast domain unless VLANs
- Router: separates broadcast domains

---

43. Hub vs Switch vs Router

Hub

- Layer 1
- sends data to all
- no address understanding

Switch

- Layer 2
- uses MAC address
- forwards in same network

Router

- Layer 3
- uses IP address
- connects different networks

---

44. Bridge vs Switch

Bridge

- older device
- fewer ports
- connects LAN segments
- uses MAC address

Switch

- advanced multiport bridge
- more ports
- faster and smarter

---

45. MAC Address vs IP Address

MAC Address

- physical address
- fixed to NIC
- used in local network
- Data Link Layer

IP Address

- logical address
- configurable
- used between networks
- Network Layer

---

46. Real-Life Analogy

Think of sending a parcel.

IP address

It is like the **city + building address** of the final destination.

MAC address

It is like the **door/person identifier for the immediate local delivery**.

Router

It is like the courier center deciding the route between cities.

Switch

It is like a local building guard directing parcel to the correct flat.

This analogy is useful in interviews.

---

47. End-to-End Communication Summary

When data is sent:

- Physical layer transmits bits
- Data link layer creates frames and uses MAC
- Network layer creates packets and uses IP

At sender:

- data becomes segment
- segment becomes packet
- packet becomes frame
- frame becomes bits

At receiver:

- bits become frame
- frame becomes packet
- packet becomes segment
- segment becomes data

---

48. Why These Layers Matter

Without the Physical Layer:

- no actual signal transmission

Without the Data Link Layer:

- no reliable local delivery
- no MAC-based transfer

Without the Network Layer:

- no communication between different networks
- no internet routing

These layers together form the backbone of networking